

Metric Spaces

A metric space (X, d) is a nonempty set with a metric $d: X \times X \rightarrow \mathbb{R}$ such that for any points $x, y, z \in X$,

- (1) $d(x, y) \geq 0$
- (2) $d(x, y) = 0 \iff x = y$
- (3) $d(x, y) = d(y, x)$
- (4) $d(x, y) \leq d(x, z) + d(z, y)$.

Open ball is $B(x_0, r) = \{x \in X : d(x_0, x) < r\}$
Closed ball is $B[x_0, r] = \{x \in X : d(x_0, x) \leq r\}$
IBB 13.5.8

True or false: In any metric space, $\overline{B(x_0, r)} = B[x_0, r]$.

Sol. False. Consider any space for example $\mathbb{R}$ with the discrete metric. $B(0, 1) = \{0\}$ and $B[0, 1] = \mathbb{R}$. Recall that

$$\overline{B(x_0, r)} = B(x_0, r) \cup \partial B(x_0, r).$$

What are the boundary points of $B(x_0, r)$?
If $a \in \mathbb{R}$ is a boundary point then certainly $B(a, 1)$ should contain a point of $B(x_0, r)$ and a point of $B(x_0, r)^c$. But $B(a, 1) = \{a\}$ only contains 1 point, so this is impossible. So the boundary is empty, and $\overline{B(0, r)} = B(0, r) \neq B[0, r]$. /E

Problem A Consider $(\mathbb{N}, d_{\mathbb{N}})$ where d is the Euclidean metric on $\mathbb{R}$. Let $E \subseteq \mathbb{N}$ be nonempty. Find

- a) E°
- b) ∂E
- c) $\bar{E}$
- d) E_{iso}

Sol. a) $E^\circ = E$. For any $x \in \mathbb{N}$, consider $B(x, 1) \subseteq \mathbb{N}$.

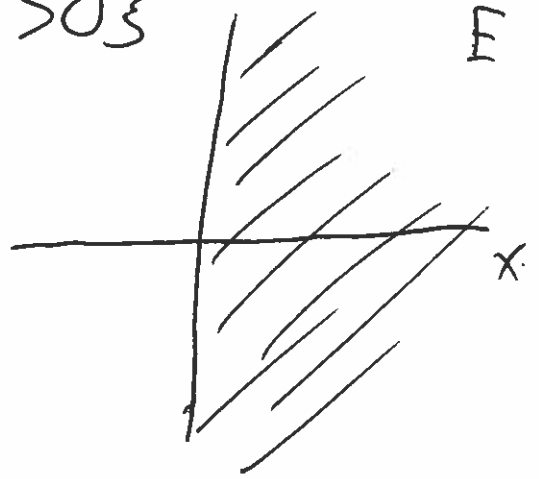
b) $\partial E = \emptyset$. For any $a \in \mathbb{N}$, $B(a, 1) = \{a\}$, so $B(a, 1)$ does not contain a point of both E and E^c , so $a \notin \partial E$.

c) $\bar{E} = E$. We have $\bar{E} = E \cup \partial E = E \cup \emptyset = E$.

d) $E_{\text{iso}} = E$. For any point $a \in E$, $B(a, 1) = \{a\}$.

Problem B Consider $\mathbb{R}^2$ with the discrete metric.
 Let $E = \{(x, y) \in \mathbb{R}^2 \mid x > 0\}$

Find each of these sets:



- a) E°
- b) ∂E
- c) $\bar{E}$
- d) E_{isol}
- e) E'

Sol. a) $E^\circ = E$. For any $x \in E$, $B(x, 1) = \{x\} \subseteq E$.

b) $\partial E = \emptyset$. For any $x \in \mathbb{R}^2$, $B(x, 1) = \{x\}$ does not contain points in E and E^c .

c) $\bar{E} = E$. $\bar{E} = E \cup \partial E = E$.

d) $E_{\text{isol}} = E$. For any $x \in E$, $B(x, 1) = \{x\}$.

e) $E' = \emptyset$. For any $x \in \mathbb{R}^2$, $B(x, 1) = \{x\}$, so $B(x, 1) \cap \{x\}^c = \emptyset$ does not contain a point of E .

Warmup: TRB 13.4.8 If $\{x_n\}$ and $\{y_n\}$ are ~~convergent~~ sequences in (X, d) and $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$, show that they are both convergent or both divergent.

Sol. Towards a contradiction, suppose WLOG that $\{x_n\}$ is convergent with $x_n \rightarrow x$. We will show that $\{y_n\}$ is also convergent with $y_n \rightarrow x$.

For any n , we have $d(x, y_n) \leq d(x, x_n) + d(x_n, y_n)$. $\textcircled{A}$

Since $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$ and $x_n \rightarrow x$, then

for any $\varepsilon > 0$, $\exists M_1, M_2 \in \mathbb{N}$ such that

note def of lim in $\mathbb{R} \rightarrow$ $|d(x_n, y_n) - 0| < \frac{\varepsilon}{2}$ for all $n \geq M_1$
and $d(x_n, x) < \frac{\varepsilon}{2}$ for all $n \geq M_2$

So let $K = \max\{M_1, M_2\}$. Then for all $n \geq K$
we have $d(x, y_n) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ by $\textcircled{A}$.

TBB 13.4.7 If $\{x_n\}$ and $\{y_n\}$ are convergent sequences in (X, d) , show that $\lim_{n \rightarrow \infty} d(x_n, y_n)$ exists.

Sol. If $x_n \rightarrow x$ and $y_n \rightarrow y$, we expect that $\lim_{n \rightarrow \infty} d(x_n, y_n) = d(x, y)$.

So we want to show: $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t.
 $|d(x_n, y_n) - d(x, y)| < \varepsilon$ for all $n \geq N$.

$$\begin{aligned} \text{We have } d(x_n, y_n) &\leq d(x_n, x) + d(x, y_n) \\ &\leq d(x_n, x) + d(y, y_n) + d(x, y) \end{aligned}$$

$$\text{so } d(x_n, y_n) - d(x, y) \leq d(x_n, x) + d(y, y_n)$$

On the other hand,

$$\begin{aligned} d(x, y) &\leq d(x_n, x) + d(x_n, y) \\ &\leq d(x_n, x) + d(y, y_n) + d(x_n, y_n) \end{aligned}$$

$$\text{so } d(x, y) - d(x_n, y_n) \leq d(x_n, x) + d(y, y_n).$$

It follows that $|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y)$.

So let $\varepsilon > 0$ be given, and choose M_1, M_2 such that $d(x_n, x) < \frac{\varepsilon}{2}$ for all $n \geq M_1$, and $d(y_n, y) < \frac{\varepsilon}{2}$ for all $n \geq M_2$.

Then for all $n \geq N = \max\{M_1, M_2\}$, we have

$$|d(x_n, y_n) - d(x, y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \text{ as desired.}$$

1BB 13.S.13 Consider the set $C[0,1]$ with the two metrics:

$$D_1(f, g) = \max_{0 \leq t \leq 1} |f(t) - g(t)|$$

$$D_2(f, g) = \int_0^1 |f(t) - g(t)| dt.$$

Let B_1 be an open ball in $(C[0,1], D_1)$.
Let B_2 be an open ball in $(C[0,1], D_2)$.

a) Is B_1 open in $(C[0,1], D_2)$?

b) Is B_2 open in $(C[0,1], D_1)$?

So. The first question to ask: how do D_1 and D_2 relate to each other?

$$D_1(f, g) = \max_{0 \leq t \leq 1} |f(t) - g(t)| < r$$

$$\Rightarrow \max_{0 \leq t \leq 1} |f(t) - g(t)| = q \text{ for some } q < r \text{ (EVT)}$$

$$\Rightarrow D_2(f, g) = \int_0^1 |f(t) - g(t)| \leq q(1-0) < r.$$

So for any $r > 0$, $B_{D_1}(f, r) \subseteq B_{D_2}(f, r)$. (★)

Is this useful for a) or b)?

Proof of b): Fix $f \in C[0,1]$, $r > 0$ and

$$B_2 := B_{D_2}(f, r) = \{g \in C[0,1] \mid D_2(f, g) < r\}$$

Since B_2 is open wrt D_2 (ex.) then each $g \in B_2$ is an interior point: $\exists s > 0$ st. $B_{D_2}(g, s) \subseteq B_2$.
We must show that g is also an interior point with respect to D_1 .

But $B_{D_1}(g, s) \subseteq B_{D_2}(g, s) \subseteq B_2$ by (*).

So B_2 is open in $(C[0,1], D_1)$.

What about a)? Consider the open ball

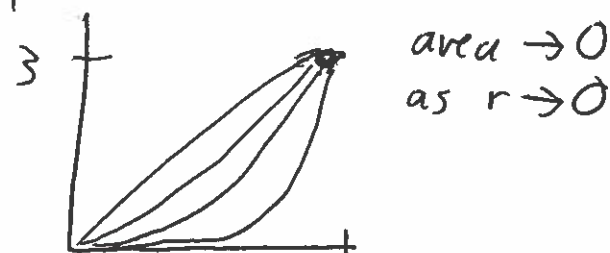
$B_1 = B_{D_1}(0, 3)$. Claim: B_1 is not open wrt. D_2 , as $f \equiv 0$ is not an interior point, but $f \in B_1$.

It suffices to show that for every $r > 0$,

$B_{D_2}(0, r) \not\subseteq B_1$. So for each $r > 0$, need to

find $h \in B_{D_2}(0, r)$ with $h \notin B_1$.

Idea: Choose h such that $\max_{0 \leq t \leq 1} |h(t) - 0| = 3$, but $\int_0^1 |h(t) - 0| dt < r$.



Let $h(t) = 3t^{3/r}$.

Observe that

$$\int_0^1 |h(t) - 0| dt = 3 \left. \frac{t^{3/r+1}}{3/r+1} \right|_0^1 = \frac{3}{\frac{3}{r}+1} \leq \frac{3}{\frac{1}{3}} = r$$

so $h \in B_{D_2}(0, r)$.

However, $\max_{0 \leq t \leq 1} |h(t) - 0| = 3$ so $h \notin B_1 = B_{D_1}(0, 3)$.

So it is not the case that B_1 is nec. open in $(C[0,1], D_2)$. ▣

TBB 13.4.3 Recall that $l_2(\mathbb{R})$ is the set of square-summable sequences:

$$l_2(\mathbb{R}) = \{ (x_1, x_2, \dots) \mid x_k \in \mathbb{R}, \sum_{k=1}^{\infty} x_k^2 < \infty \}.$$

$$\text{If we define } d_2(x, y) = \left(\sum_{k=1}^{\infty} (x_k - y_k)^2 \right)^{1/2}$$

where $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$, then $(l_2(\mathbb{R}), d_2)$ is a metric space.

Q: Let $x^{(n)} = (x_1^{(n)}, x_2^{(n)}, \dots)$ be a sequence of points in $l_2(\mathbb{R})$ and $y = (y_1, y_2, \dots) \in l_2(\mathbb{R})$. Is it true that

$$x^{(n)} \rightarrow y \text{ in } l_2(\mathbb{R})$$

if and only if

$$\lim_{n \rightarrow \infty} x_k^{(n)} = y_k \text{ for each } k=1, 2, \dots?$$

Sol. ($\Rightarrow$) Suppose that $x^{(n)} \rightarrow y$. Then for any $\varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\sum_{k=1}^{\infty} (x_k^{(n)} - y_k)^2 < \varepsilon^2$ whenever $n \geq N$.

But for each fixed $k=1, 2, \dots$,

$$(x_k^{(n)} - y_k)^2 \leq \sum_{i=1}^{\infty} (x_i^{(n)} - y_i)^2 < \varepsilon^2 \text{ whenever } n \geq N$$

So $|x_k^{(n)} - y_k| = \sqrt{(x_k^{(n)} - y_k)^2} < \varepsilon$ whenever $n \geq N$,
i.e., $\lim_{n \rightarrow \infty} x_k^{(n)} = y_k$.

($\Leftarrow$) What about the converse?

Consider the case that $y = (0, 0, \dots)$.

Consider the sequence of points

$$x^{(n)} = (0, \dots, 0, \underset{x^{(n)}_n}{1}, 0, \dots)$$

$$\text{where } x^{(n)} = (x^{(n)}_1, x^{(n)}_2, \dots)$$

$$\text{and } x^{(n)}_k = \begin{cases} 1 & \text{if } k=n \\ 0 & \text{otherwise} \end{cases}$$

For each $k=1, 2, \dots$, we have $\lim_{n \rightarrow \infty} x^{(n)}_k = 0$.

However, for any n , we have

$$\begin{aligned} \sqrt{\sum_{k=1}^{\infty} (x^{(n)}_k - y_k)^2} &= \sqrt{\sum_{k=1}^{\infty} (x^{(n)}_k)^2} \\ &= \sqrt{\sum_{k=1}^{\infty} (x^{(n)}_k)^2} = 1 \end{aligned}$$

So in particular, for ~~$\epsilon = 1/2$~~ $\epsilon = 1/2$, there is no n such that

$$d_2(x^{(n)}, y) < \frac{1}{2}, \text{ so } x^{(n)} \not\rightarrow y,$$

the converse must be false!



TBB 13.6.17 Consider the metric space $(C[0,1], d)$ where $d(f,g) = \int_0^1 |f-g|$. Define the functional $T: C[0,1] \rightarrow \mathbb{R}$ by $T(f) = \int_0^1 f(t) dt$. Is T continuous?

Sol. Yes. Let $f_n \rightarrow f$ in $(C[0,1], d)$, we must show that $T(f_n) \rightarrow T(f)$ in $(\mathbb{R}, \text{standard})$. Let $\epsilon > 0$. $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N$,

$$d(f_n, f) = \int_0^1 |f_n(t) - f(t)| dt < \epsilon$$

But for any $n \in \mathbb{N}$, we have

$$\begin{aligned} |T(f_n) - T(f)| &= \left| \int_0^1 f_n(t) dt - \int_0^1 f(t) dt \right| \\ &= \left| \int_0^1 (f_n(t) - f(t)) dt \right| \\ &\leq \int_0^1 |f_n(t) - f(t)| dt \quad (\text{exercise}) \end{aligned}$$

so $|T(f_n) - T(f)| \leq d(f_n, f) < \epsilon$ for all $n \geq N$.

Hence T is continuous at f , and hence on $(C[0,1], d)$ since f is arbitrary. 

TBB 13.S.4 Let E be a closed set in (X, d) .

Let $x \notin E$. Show that

$$\inf \{d(x, y) : y \in E\} > 0.$$

If E and F are disjoint ^{closed} sets, is it true that $\inf \{d(x, y) : x \in E, y \in F\} > 0$?

Sol. Since E is closed, x cannot be a limit point, as otherwise $x \in E$. So $\exists r > 0$ such that $B(x, r) \cap E = \emptyset$, as $x \notin E$. Hence for any $y \in E$, we must have $d(x, y) \geq r$. So

$$\inf \{d(x, y) : y \in E\} \geq r > 0 \text{ as desired.}$$

Answer is NO. Consider the metric space $\mathbb{R} \setminus \{0\}$ with the ~~usual~~ standard metric. Let

$E = (-\infty, 0)$ and $F = (0, \infty)$. Then E and F are disjoint. Moreover E and F are closed as they contain their limit points (note $0 \in \mathbb{R} \setminus \{0\}$).

However, if we let $x_n = \frac{1}{n}$ and $y_n = -\frac{1}{n}$ for each $n \geq 1$, then $d(x_n, y_n) = \frac{2}{n}$, so since

$\inf \leq \frac{2}{n}$ for all n , we have $\inf = 0$. 

TBB 13.3.5 Let $R[0,1]$ be the set of integrable functions on $[0,1]$. Let

$$d(f,g) = \int_0^1 |f(t) - g(t)| dt.$$

Is $(R[0,1], d)$ a metric space?

Sol. Non-negativity, symmetry are obviously satisfied. ~~$d(f,g) \geq 0$~~ Triangle inequality also satisfied via triangle inequality on $\mathbb{R}$. We also need that $d(f,g) = 0 \iff f = g$.

However, consider $f \equiv 0$

$$g(x) = \begin{cases} 1 & x = 1/2 \\ 0 & x \neq 1/2 \end{cases}$$

Both f and g are integrable and $|f(x) - g(x)| = g(x)$ for all $x \in [0,1]$.

$$\text{But then } d(f,g) = \int_0^1 |f(t) - g(t)| = \int_0^1 g(t) dt = 0$$

(using our knowledge of integrals).

However, $f \neq g$, so d cannot be a metric on $R[0,1]$. 

TBR 13.5.15 (The Hilbert cube)

Consider the following subsets of ℓ_2 :

$$H = \{(x_1, x_2, \dots) \in \ell_2 \mid |x_i| \leq 1/i\}$$

$$G = \{(x_1, x_2, \dots) \in \ell_2 \mid |x_i| < 1/i\}$$

H is called the Hilbert cube.

a) Is H closed in ℓ_2 ?

b) Is G open in ℓ_2 ?

Sol. ^{a)} Let's show H is closed. Let $y = (y_1, y_2, \dots)$ be a limit point of H . We must show $y \in H$. We know there exists a sequence of points $x^{(n)} = (x_1^{(n)}, x_2^{(n)}, \dots) \in H$ s.t. $x^{(n)} \xrightarrow{n \rightarrow \infty} y$.

From last week, we know this implies that for each k , $\lim_{n \rightarrow \infty} x_k^{(n)} = y_k$. So since each $x^{(n)} \in H$,

we have that for each k

$$|x_k^{(n)}| \leq 1/k \Rightarrow -\frac{1}{k} \leq x_k^{(n)} \leq \frac{1}{k}$$

$$\Rightarrow -\frac{1}{k} \leq \lim_{n \rightarrow \infty} x_k^{(n)} \leq \frac{1}{k} \quad \text{taking limits}$$

$$\Rightarrow -\frac{1}{k} \leq y_k \leq \frac{1}{k}$$

$$\Rightarrow |y_k| \leq 1/k$$

So $y \in H$ as desired.

b) Somewhat surprisingly, G is not open. As with last week, let $y = (0, 0, \dots) \in G$. We will show that y is not an interior point of G . Recall that

$$B(0, r) = \{(x_1, x_2, \dots) \in \ell_2 \mid \sqrt{\sum_{i=1}^{\infty} x_i^2} < r\}$$

We need to show that for any $r > 0$,

$B(0, r) \not\subset G$. In fact, it suffices to show that for all $0 < r < 1$, $B(0, r) \not\subset G$. In fact, it's enough to show that $B(0, \frac{1}{k}) \not\subset G$ for each ~~$k \geq 2$~~ $k \geq 2$.

Naïve idea: let $x = (0, 0, \dots, 0, \overset{x_k}{\frac{1}{k}}, \frac{1}{k^2}, \frac{1}{k^3}, \dots)$ then $x \notin G$ since $|x_k| \not\leq \frac{1}{k}$. But sadly, $\sqrt{\sum_{i=1}^{\infty} x_i^2} = \sqrt{\sum_{i=1}^{\infty} (\frac{1}{k^i})^2}$ is not $< \frac{1}{k}$, so $x \notin B(0, \frac{1}{k})$.

New idea: let $x = (0, 0, \dots, 0, \overset{x_{k^2}}{\frac{1}{k^2}}, (\frac{1}{k^2})^2, (\frac{1}{k^2})^3, \dots)$.

Now, we can compute

$$\begin{aligned} \sum_{i=1}^{\infty} x_i^2 &= \sum_{i=1}^{\infty} (\frac{1}{k^2})^i = \sum_{i=0}^{\infty} (\frac{1}{k^2}) (\frac{1}{k^2})^i \\ &= \frac{\frac{1}{k^2}}{1 - \frac{1}{k^2}} = \frac{1}{k^2 - 1} < \frac{1}{k} \quad \text{since } k \geq 2 \end{aligned}$$

So $x \in B(0, \frac{1}{k})$, but as before $|x_{k^2}| \not\leq \frac{1}{k^2}$, so $x \notin G$. So G is not open, as $(0, 0, \dots)$ is not an interior point.